

A.1 Proof of Lemma 3.1

Proof. Recall that $m_0 \in C^r(\mathcal{Z})$, where $\mathcal{Z} \subset \mathbb{R}^q$ is compact and $r \geq 2$. Let $\mathcal{M}_{K_n}$ denote the spline sieve space spanned by B-spline basis functions $\{B_1, \dots, B_{K_n}\}$ of order at least r , constructed over a quasi-uniform partition of $\mathcal{Z}$.

For notational clarity, consider first the case $q = 1$, i.e., $\mathcal{Z} = [0, 1]$. The multivariate case follows by tensor-product spline constructions with analogous rates. Let the interval $[0, 1]$ be partitioned into K_n subintervals with knots $0 = t_0 < t_1 < \dots < t_{K_n} = 1$, where the mesh size satisfies

$$h_n := \max_{1 \leq j \leq K_n} (t_j - t_{j-1}) \asymp K_n^{-1},$$

where $\asymp$ stands for *asymptotic equivalence*. Quasi-uniformity ensures that $h_n \leq CK_n^{-1}$ for some constant $C > 0$.

Since $m_0 \in C^r([0, 1])$, for each subinterval $[t_{j-1}, t_j]$ we may approximate m_0 by its Taylor polynomial of degree $r - 1$ around a point in the interval. By standard Taylor expansion with remainder, for any $z \in [t_{j-1}, t_j]$,

$$m_0(z) = P_{j,r-1}(z) + R_j(z),$$

where $P_{j,r-1}$ is a polynomial of degree $r - 1$ and the remainder satisfies $|R_j(z)| \leq Ch_n^r$, with C depending only on the supremum norm of the r -th derivative of m_0 . Hence, there exists a piecewise polynomial function p_n of degree $r - 1$ such that

$$\|m_0 - p_n\|_\infty \leq Ch_n^r.$$

The spline space $\mathcal{M}_{K_n}$ contains all piecewise polynomials of degree $r - 1$ with continuity constraints up to order $r - 2$. Classical results in spline approximation theory (see, e.g., de Boor (2001)[5]) guarantee that for any sufficiently smooth function, there exists a spline $m_{K_n} \in \mathcal{M}_{K_n}$ such that

$$\|p_n - m_{K_n}\|_\infty \leq Ch_n^r.$$

Combining with the previous step yields

$$\|m_0 - m_{K_n}\|_\infty \leq \|m_0 - p_n\|_\infty + \|p_n - m_{K_n}\|_\infty \leq Ch_n^r.$$

Since $h_n \asymp K_n^{-1}$, we obtain

$$\|m_0 - m_{K_n}\|_\infty \leq CK_n^{-r}.$$

For $q > 1$, the result follows by considering tensor-product B-spline bases over a quasi-uniform grid on $\mathcal{Z} \subset \mathbb{R}^q$. Under the assumption $m_0 \in C^r(\mathcal{Z})$, standard multivariate spline approximation results yield the same rate (up to constants depending on q), namely

$$\inf_{m \in \mathcal{M}_{K_n}} \|m_0 - m\|_\infty \leq CK_n^{-r}.$$

Thus, there exists a spline function $m_{K_n} \in \mathcal{M}_{K_n}$ satisfying

$$\|m_0 - m_{K_n}\|_\infty = O(K_n^{-r}),$$

which completes the proof. $\square$

A.2 Proof of Lemma 3.2

Proof. Recall that the weights are defined as

$$w_i(\beta, m) = \frac{1}{C(X_i, Z_i; \beta, m)},$$

where $C(x, z; \beta, m) = \int_{a_Y}^{b_Y} f_\varepsilon(y - m(z) - g(x; \beta)) dy$.

By Assumption 4 (Truncation Stability), there exists a constant $c > 0$ such that

$$\inf_{x, z} C(x, z; \beta, m) \geq c$$

for all (β, m) in a neighborhood of (β_0, m_0) . In particular, this implies that for each observation (X_i, Z_i) ,

$$C(X_i, Z_i; \beta, m) \geq c \quad \text{almost surely.}$$

Using the definition of $w_i(\beta, m)$, we obtain

$$w_i(\beta, m) = \frac{1}{C(X_i, Z_i; \beta, m)} \leq \frac{1}{c} \quad \text{for all } i = 1, \dots, n.$$

Thus, it can be concluded that $\sup_{1 \leq i \leq n} w_i(\beta, m) \leq \frac{1}{c}$.

Since the upper bound $1/c$ is a deterministic constant independent of n , it follows immediately that

$$\sup_{1 \leq i \leq n} w_i(\beta, m) = O_p(1).$$

Because $(\hat{\beta}, \hat{m}) \rightarrow_p (\beta_0, m_0)$ and the truncation stability condition holds in a neighborhood of (β_0, m_0) , the same bound applies when (β, m) is replaced by $(\hat{\beta}, \hat{m})$ with probability tending to one. Hence the weights remain uniformly bounded in probability along the estimator sequence.

Therefore, we obtain that $\sup_{1 \leq i \leq n} w_i(\beta, m) = O_p(1)$, which completes the proof. $\square$

A.3 Proof of Lemma 3.3

Proof. Recall that the log-likelihood is given by

$$\ell_n(\beta, \theta) = \sum_{i=1}^n \log f_\varepsilon(Y_i - m(Z_i) - g(X_i; \beta)) - \sum_{i=1}^n \log C(X_i, Z_i; \beta, m),$$

where $m(z) = \sum_{k=1}^{K_n} \theta_k B_k(z)$. Let $S_{\theta, n} = \nabla_\theta \ell_n(\beta, \theta)$ denote the score vector with respect to θ .

Differentiating $\ell_n(\beta, \theta)$ with respect to θ_k , we obtain

$$\frac{\partial \ell_n}{\partial \theta_k} = \sum_{i=1}^n \left[-\dot{\ell}_\varepsilon(R_i(\beta, \theta)) B_k(Z_i) + \frac{\partial C(X_i, Z_i; \beta, m) / \partial \theta_k}{C(X_i, Z_i; \beta, m)} \right],$$

where $R_i(\beta, \theta) = Y_i - m(Z_i) - g(X_i; \beta)$ and $\dot{\ell}_\varepsilon(u) = \frac{d}{du} \log f_\varepsilon(u)$. Thus, one can express that $S_{\theta,n} = \sum_{i=1}^n \psi_i(\beta, \theta)$, where each $\psi_i(\beta, \theta)$ is a K_n -dimensional vector with k -th component

$$\psi_{i,k} = -\dot{\ell}_\varepsilon(R_i(\beta, \theta)) B_k(Z_i) + \frac{\partial C(X_i, Z_i; \beta, m) / \partial \theta_k}{C(X_i, Z_i; \beta, m)}.$$

By Assumption 5, $\dot{\ell}_\varepsilon(\cdot)$ is bounded. Since the spline basis functions are uniformly bounded on the compact domain $\mathcal{Z}$, there exists a constant C_1 such that $|\dot{\ell}_\varepsilon(R_i) B_k(Z_i)| \leq C_1$. Next, consider the second term. By differentiation under the integral sign,

$$\frac{\partial C(x, z; \beta, m)}{\partial \theta_k} = - \int_{a_Y}^{b_Y} f'_\varepsilon(y - m(z) - g(x; \beta)) B_k(z) dy.$$

Using boundedness of f'_ε and B_k , we obtain $\left| \frac{\partial C(x, z; \beta, m)}{\partial \theta_k} \right| \leq C_2$.

Under the truncation stability assumption, $C(x, z; \beta, m) \geq c > 0$, hence

$$\left| \frac{\partial C(x, z; \beta, m) / \partial \theta_k}{C(x, z; \beta, m)} \right| \leq \frac{C_2}{c}.$$

Therefore, each component $\psi_{i,k}$ is uniformly bounded, i.e., $|\psi_{i,k}| \leq C_3$. Because the observations are independent, we have $\text{Var}(S_{\theta,n}) = \sum_{i=1}^n \text{Var}(\psi_i)$. Each ψ_i is a K_n -dimensional vector with bounded components, so its covariance matrix has operator norm bounded by CK_n for some constant C . Hence,

$$\text{Var}(S_{\theta,n}) = O(nK_n).$$

Using the identity $\mathbb{E}\|S_{\theta,n}\|^2 = \text{trace}(\text{Var}(S_{\theta,n})) + \|\mathbb{E}S_{\theta,n}\|^2$, and noting that $\|\mathbb{E}S_{\theta,n}\| = O(n)$ but is dominated by variance after centering arguments (or working locally around the true parameter), we obtain

$$\mathbb{E}\|S_{\theta,n}\|^2 = O(nK_n).$$

Applying Markov's inequality (or Chebyshev's inequality) yields $\|S_{\theta,n}\| = O_p(\sqrt{nK_n})$.

The factor $\sqrt{n}$ reflects aggregation over independent observations, while the factor $\sqrt{K_n}$ reflects the dimensionality of the spline coefficient vector. Thus the score behaves like a K_n -dimensional random vector with variance proportional to n per coordinate. This completes the proof. $\square$

A.4 Proof of Theorem 3.4

Proof. Let

$$\ell_n(\beta, m) = \sum_{i=1}^n \log f_\varepsilon \{Y_i - m(Z_i) - g(X_i; \beta)\} - \sum_{i=1}^n \log C(X_i, Z_i; \beta, m)$$

denote the truncated log-likelihood. Define the normalized empirical criterion by

$$M_n(\beta, m) = \frac{1}{n} \ell_n(\beta, m).$$

Let

$$M(\beta, m) = E [\log f_\varepsilon \{Y - m(Z) - g(X; \beta)\} - \log C(X, Z; \beta, m)]$$

be the corresponding population criterion.

The main difference from the conventional untruncated partially nonlinear regression model is the presence of the normalization term $\log C(X, Z; \beta, m)$. In particular, the derivatives of the criterion with respect to β and m contain additional terms generated by this truncation correction. For the parametric component,

$$\frac{\partial M_n(\beta, m)}{\partial \beta} = \frac{1}{n} \sum_{i=1}^n \frac{\partial}{\partial \beta} \log f_\varepsilon \{Y_i - m(Z_i) - g(X_i; \beta)\} - \frac{1}{n} \sum_{i=1}^n \frac{\partial}{\partial \beta} \log C(X_i, Z_i; \beta, m).$$

Similarly, for a perturbation $m_t = m + th$ of the nonparametric component, the functional derivative contains the additional term

$$- \left. \frac{d}{dt} \log C(X, Z; \beta, m_t) \right|_{t=0}.$$

Thus, both components of the estimator are affected by the parameter-dependent truncation normalization.

We first establish uniform convergence of the empirical criterion. Write

$$M_n(\beta, m) - M(\beta, m) = \{P_n - P\} \log f_\varepsilon \{Y - m(Z) - g(X; \beta)\} - \{P_n - P\} \log C(X, Z; \beta, m),$$

where P_n denotes the empirical measure and P the corresponding population measure. Therefore,

$$\sup_{(\beta, m)} |M_n(\beta, m) - M(\beta, m)| \leq A_n + B_n,$$

where

$$A_n = \sup_{(\beta, m)} |(P_n - P) \log f_\varepsilon \{Y - m(Z) - g(X; \beta)\}|$$

and

$$B_n = \sup_{(\beta, m)} |(P_n - P) \log C(X, Z; \beta, m)|.$$

The first term A_n is controlled by the standard empirical-process arguments used for sieve M-estimation. The second term B_n is specific to the truncation-adjusted likelihood. Under the assumed regularity conditions, there exists $c_0 > 0$ such that

$$C(X, Z; \beta, m) \geq c_0$$

uniformly over the parameter space and the sieve space. Hence, $\log C(X, Z; \beta, m)$ is uniformly bounded whenever C is uniformly bounded above. Moreover, the assumed smoothness of C implies that

$$\sup_{(\beta, m)} \left\| \frac{\partial}{\partial \beta} \log C(X, Z; \beta, m) \right\| < \infty$$

and

$$\sup_{(\beta, m)} \left| \frac{d}{dt} \log C(X, Z; \beta, m_t) \right|_{t=0} < \infty.$$

Consequently, the empirical-process class generated by $\log C(X, Z; \beta, m)$ and its relevant derivatives is uniformly Glivenko–Cantelli under the stated entropy conditions. It follows that

$$B_n = o_p(1).$$

Therefore,

$$\sup_{(\beta, m)} |M_n(\beta, m) - M(\beta, m)| = o_p(1).$$

Next, by the identifiability assumption, the population criterion $M(\beta, m)$ has a unique maximizer at (β_0, m_0) . Hence, for every $\epsilon > 0$, there exists $\eta_\epsilon > 0$ such that

$$\sup_{\|(\beta, m) - (\beta_0, m_0)\| \geq \epsilon} M(\beta, m) \leq M(\beta_0, m_0) - \eta_\epsilon.$$

Together with the uniform convergence established above, this implies

$$P \left(\sup_{\|(\beta, m) - (\beta_0, m_0)\| \geq \epsilon} M_n(\beta, m) < M_n(\beta_0, m_0) \right) \rightarrow 1.$$

Therefore, the maximizer of the empirical criterion must lie in an arbitrarily small neighborhood of (β_0, m_0) with probability tending to one.

For the spline approximation, let $\mathcal{M}_n$ denote the sieve space used for approximating m_0 . Under the assumed smoothness condition, there exists $m_{0,n} \in \mathcal{M}_n$ satisfying

$$\|m_{0,n} - m_0\| = O(K_n^{-r}),$$

where K_n denotes the dimension of the spline sieve. The approximation error tends to zero under the imposed conditions on K_n . The truncation correction does not alter this approximation property because the normalization term is continuously differentiable in m and is uniformly bounded away from zero. In particular,

$$|\log C(X, Z; \beta, m_{0,n}) - \log C(X, Z; \beta, m_0)| \leq C_2 \|m_{0,n} - m_0\|$$

for some finite constant C_2 . Hence,

$$\sup_{\beta} |P \log C(X, Z; \beta, m_{0,n}) - P \log C(X, Z; \beta, m_0)| = o(1).$$

Combining the stochastic convergence of the empirical criterion, the spline approximation property, and the uniqueness of the population maximizer yields

$$\|\hat{\beta} - \beta_0\| + \|\hat{m} - m_0\| \xrightarrow{p} 0.$$

Thus, the proposed truncation-adjusted estimator is uniformly consistent.

The proof also shows explicitly where the truncation mechanism enters the consistency argument. In addition to the usual empirical-process term associated with the regression likelihood, one must control the empirical process generated by

$$\log C(X, Z; \beta, m),$$

together with its derivatives

$$\frac{\partial}{\partial \beta} \log C(X, Z; \beta, m)$$

and

$$\frac{d}{dt} \log C(X, Z; \beta, m_t).$$

The lower bound on C and the assumed smoothness and entropy conditions ensure that these additional terms are uniformly controlled. Therefore, the standard sieve M-estimation argument can be applied after explicitly accounting for the truncation-dependent normalization.

This completes the proof. $\square$

A.5 Proof of Theorem 3.5

Proof. We decompose the estimation error into the spline approximation component and the stochastic estimation component as

$$\hat{m} - m_0 = (\hat{m} - m_{K_n}) + (m_{K_n} - m_0),$$

where m_{K_n} denotes the best spline approximation to m_0 in the K_n -dimensional spline space.

By the spline approximation result, since m_0 belongs to the r -smooth function class,

$$\|m_{K_n} - m_0\|_\infty = O(K_n^{-r}).$$

Thus, the first component represents the deterministic approximation error generated by replacing the infinite-dimensional function m_0 by its finite-dimensional spline approximation.

Let θ_{K_n} denote the coefficient vector corresponding to m_{K_n} . The second component is controlled by expanding the truncation-adjusted penalized likelihood around (β_0, θ_{K_n}) . The score with respect to the spline coefficients contains not only the usual likelihood score but also the additional contribution generated by the truncation-normalization constant. In particular,

$$S_{\theta,n}(\beta, \theta) = \frac{\partial \ell_n(\beta, \theta)}{\partial \theta} - 2\lambda_n \Omega_n \theta,$$

where the derivative of the truncation correction contains terms of the form

$$-\frac{D_\theta C(X_i, Z_i; \beta, m_\theta)}{C(X_i, Z_i; \beta, m_\theta)}.$$

The lower bound on the truncation probability therefore ensures that these additional score contributions remain uniformly bounded.

At the penalized estimator $(\hat{\beta}, \hat{\theta})$, the first-order condition gives

$$0 = S_{\theta,n}(\beta_0, \theta_{K_n}) - H_{\theta\theta,n}(\tilde{\beta}, \tilde{\theta})(\hat{\theta} - \theta_{K_n}) - H_{\theta\beta,n}(\tilde{\beta}, \tilde{\theta})(\hat{\beta} - \beta_0) - 2\lambda_n \Omega_n \hat{\theta} + R_n,$$

where $(\tilde{\beta}, \tilde{\theta})$ lies between (β_0, θ_{K_n}) and $(\hat{\beta}, \hat{\theta})$, and R_n contains the higher-order remainder terms.

By Lemma 3.3, the spline score satisfies

$$\|S_{\theta,n}(\beta_0, \theta_{K_n})\| = O_p(\sqrt{nK_n}),$$

because the score contains K_n spline coordinates and the truncation probability is bounded away from zero. Moreover, the truncation-dependent terms satisfy the same stochastic order under the stated regularity conditions. In particular, the empirical-process contribution associated with

$$\frac{D_{\theta}C}{C}$$

is uniformly controlled by the assumed lower bound on C and the boundedness conditions imposed on its derivatives.

The spline Hessian admits the expansion

$$H_{\theta\theta,n}(\tilde{\beta}, \tilde{\theta}) = n\mathcal{I}_{\theta\theta} + o_p(n),$$

where $\mathcal{I}_{\theta\theta}$ is the population information operator restricted to the sieve space. The truncation adjustment changes this information operator through the derivatives of $\log C$, but under the truncation-stability condition

$$\inf_{x,z} C(x, z; \beta, m) \geq c_C > 0,$$

together with the assumed smoothness and boundedness conditions, $\mathcal{I}_{\theta\theta}$ remains uniformly positive definite. Hence, for some constant $c > 0$,

$$\lambda_{\min}\{H_{\theta\theta,n}\} \geq cn$$

with probability tending to one. Consequently,

$$\|H_{\theta\theta,n}^{-1}\| = O_p(n^{-1}).$$

Ignoring terms that are of smaller order under the stated rate conditions, the preceding expansion gives

$$\hat{\theta} - \theta_{K_n} = H_{\theta\theta,n}^{-1} S_{\theta,n}(\beta_0, \theta_{K_n}) + O_p(\lambda_n) + o_p(\|\hat{\theta} - \theta_{K_n}\|).$$

Therefore,

$$\|\hat{\theta} - \theta_{K_n}\| = O_p\left(\frac{\sqrt{nK_n}}{n}\right) + O_p(\lambda_n) = O_p\left(\sqrt{\frac{K_n}{n}} + \lambda_n\right).$$

Under the assumed conditions on λ_n , the penalty contribution is asymptotically negligible relative to the leading stochastic term. Hence,

$$\|\hat{\theta} - \theta_{K_n}\| = O_p \left(\sqrt{\frac{K_n}{n}} \right).$$

Since the spline basis is uniformly bounded, the corresponding function-level estimation error satisfies

$$\|\hat{m} - m_{K_n}\|_{L_2} = O_p \left(\sqrt{\frac{K_n}{n}} \right).$$

Combining this stochastic estimation error with the approximation error gives

$$\begin{aligned} \|\hat{m} - m_0\|_{L_2} &\leq \|\hat{m} - m_{K_n}\|_{L_2} + \|m_{K_n} - m_0\|_{L_2} \\ &= O_p \left(\sqrt{\frac{K_n}{n}} \right) + O(K_n^{-r}). \end{aligned}$$

Thus,

$$\boxed{\|\hat{m} - m_0\|_{L_2} = O_p \left(K_n^{-r} + \sqrt{\frac{K_n}{n}} \right).}$$

The result shows explicitly the two principal sources of estimation error. The term K_n^{-r} is the deterministic spline approximation error, whereas $\sqrt{K_n/n}$ is the stochastic error associated with estimating an increasing number of spline coefficients. The truncation-dependent normalization does not introduce a separate dominant rate under the assumed condition that C is uniformly bounded away from zero; rather, it modifies the score and information through the derivatives of $\log C$. This condition is therefore essential for retaining the stated convergence rate. □

A.6 Proof of Theorem 3.6

Proof. Let

$$S_{\beta,n}(\beta, m) = \frac{\partial \ell_n(\beta, m)}{\partial \beta}$$

denote the score function with respect to the finite-dimensional parameter β . From the truncated log-likelihood in (2.4), we have

$$S_{\beta,n}(\beta, m) = \sum_{i=1}^n \psi_{\beta,i}(\beta, m),$$

where

$$\psi_{\beta,i}(\beta, m) = \frac{\partial}{\partial \beta} \log f_\varepsilon \{Y_i - m(Z_i) - g(X_i; \beta)\} - \frac{\partial}{\partial \beta} \log C(X_i, Z_i; \beta, m).$$

The second term is the additional contribution generated by the truncation mechanism. In the absence of truncation, $C(X_i, Z_i; \beta, m) \equiv 1$, and hence this term is identically zero.

Let (β_0, m_0) denote the true parameter-function pair and define

$$\psi_{\beta,i}^0 = \psi_{\beta,i}(\beta_0, m_0).$$

Under the assumed regularity conditions, the conditional likelihood is correctly specified and therefore

$$E(\psi_{\beta,i}^0) = 0.$$

Moreover, the assumptions on the truncation probabilities imply that there exists a constant $c_0 > 0$ such that

$$C(X, Z; \beta, m) \geq c_0$$

uniformly over the relevant parameter and sieve spaces. Consequently, the derivatives of $\log C(X, Z; \beta, m)$ are uniformly bounded whenever the corresponding derivatives of C are bounded. In particular,

$$\sup_{(\beta, m)} \left\| \frac{\partial}{\partial \beta} \log C(X, Z; \beta, m) \right\| < \infty,$$

and analogous bounds hold for the second-order derivatives required below.

We first consider the effect of estimating the nuisance function m . By Theorem 3.5,

$$\|\widehat{m} - m_0\| = O_p(n^{-r/(2r+1)}).$$

For $r \geq 2$,

$$n^{-r/(2r+1)} = o(n^{-1/4}),$$

and hence

$$\|\widehat{m} - m_0\| = o_p(n^{-1/4}).$$

This rate is sufficient for the nuisance contribution to the $\sqrt{n}$ -scaled parametric expansion to be asymptotically negligible after the corresponding first-order orthogonality and second-order remainder terms are taken into account.

To make this explicit, consider a Taylor expansion of the score around (β_0, m_0) . We have

$$0 = S_{\beta,n}(\widehat{\beta}, \widehat{m}) = S_{\beta,n}(\beta_0, m_0) + \frac{\partial S_{\beta,n}}{\partial \beta} \Big|_{(\beta_0, m_0)} (\widehat{\beta} - \beta_0) + \mathcal{R}_{\beta,n},$$

where $\mathcal{R}_{\beta,n}$ contains the terms generated by estimating m and the higher-order Taylor remainder. Equivalently,

$$0 = S_{\beta,n}(\beta_0, m_0) - I_n(\beta_0, m_0)(\widehat{\beta} - \beta_0) + \mathcal{R}_{\beta,n},$$

where

$$I_n(\beta, m) = -\frac{\partial S_{\beta,n}(\beta, m)}{\partial \beta}$$

is the observed information matrix.

The truncation correction contributes to this information matrix through

$$-\frac{\partial^2}{\partial\beta\partial\beta^\top}\log C(X_i, Z_i; \beta, m).$$

Thus,

$$I_n(\beta, m) = -\sum_{i=1}^n \frac{\partial^2}{\partial\beta\partial\beta^\top} \log f_\varepsilon \{Y_i - m(Z_i) - g(X_i; \beta)\} + \sum_{i=1}^n \frac{\partial^2}{\partial\beta\partial\beta^\top} \log C(X_i, Z_i; \beta, m).$$

The second summation is absent under ordinary untruncated sampling.

The nuisance-function contribution also contains a truncation-specific component. For a perturbation $m_t = m + th$, the functional derivative of the score contains

$$-\left. \frac{\partial}{\partial\beta} \frac{d}{dt} \log C(X_i, Z_i; \beta, m_t) \right|_{t=0}.$$

Hence, uniformly in admissible directions h ,

$$\left\| \frac{\partial S_{\beta,n}}{\partial m}(\beta_0, m_0)[h] \right\| \leq C_1 \sum_{i=1}^n |h(Z_i)|$$

for some finite constant C_1 , under the assumed boundedness conditions on the derivatives of C . The empirical-process terms generated by the truncation correction can therefore be controlled by the same sieve norm used in Theorem 3.5. In particular,

$$\sup_{m \in \mathcal{M}_n} \left| \left\{ (P_n - P) \left[\frac{\partial}{\partial\beta} \log C(X, Z; \beta, m) \right] \right\} \right| = o_p(n^{-1/2}),$$

under the entropy and smoothness conditions imposed on the sieve class. Similarly, the corresponding second-order truncation terms are uniformly bounded and are absorbed into the remainder $\mathcal{R}_{\beta,n}$.

Using

$$\|\widehat{m} - m_0\| = o_p(n^{-1/4}),$$

the smoothness assumptions on the likelihood and on the normalization term imply

$$\|\mathcal{R}_{\beta,n}\| = o_p(\sqrt{n}).$$

Therefore,

$$0 = S_{\beta,n}(\beta_0, m_0) - I_n(\beta_0, m_0)(\widehat{\beta} - \beta_0) + o_p(\sqrt{n}).$$

Dividing by $\sqrt{n}$ gives

$$0 = \frac{1}{\sqrt{n}} S_{\beta,n}(\beta_0, m_0) - \frac{I_n(\beta_0, m_0)}{n} \sqrt{n}(\widehat{\beta} - \beta_0) + o_p(1).$$

Next, by the uniform law of large numbers applied to the likelihood derivative class, including the class generated by the truncation normalization, we have

$$\frac{1}{n} I_n(\beta_0, m_0) \xrightarrow{p} I(\beta_0),$$

where

$$I(\beta_0) = -E \left[\frac{\partial^2}{\partial \beta \partial \beta^\top} \log f_\varepsilon \{Y - m_0(Z) - g(X; \beta_0)\} \right] + E \left[\frac{\partial^2}{\partial \beta \partial \beta^\top} \log C(X, Z; \beta_0, m_0) \right].$$

The second expectation is the additional information contribution arising from truncation. By assumption, $I(\beta_0)$ is positive definite and therefore nonsingular.

It remains to establish the limiting distribution of the score. At the true parameter values,

$$S_{\beta,n}(\beta_0, m_0) = \sum_{i=1}^n \psi_{\beta,i}^0,$$

with

$$\psi_{\beta,i}^0 = \frac{\partial}{\partial \beta} \log f_\varepsilon \{Y_i - m_0(Z_i) - g(X_i; \beta_0)\} - \frac{\partial}{\partial \beta} \log C(X_i, Z_i; \beta_0, m_0).$$

The observations are independent under the conditional truncated sampling scheme. The truncation-specific score term is uniformly bounded by the assumptions on C and its derivatives. Together with the assumed moment conditions on the ordinary likelihood score, this implies

$$E \|\psi_{\beta,i}^0\|^2 < \infty.$$

Consequently, the Lindeberg condition holds, and the Lindeberg–Feller central limit theorem gives

$$\frac{1}{\sqrt{n}} S_{\beta,n}(\beta_0, m_0) \xrightarrow{d} N\{0, \Sigma(\beta_0)\},$$

where

$$\Sigma(\beta_0) = \text{Var} [\psi_{\beta,i}^0].$$

Under the correctly specified likelihood and the information identity, $\Sigma(\beta_0) = I(\beta_0)$. More generally, if the information identity is not imposed, the limiting covariance matrix is obtained from the usual sandwich form.

Combining the score expansion, the convergence of the observed information, and the central limit theorem, we obtain

$$\sqrt{n}(\hat{\beta} - \beta_0) = I(\beta_0)^{-1} \frac{1}{\sqrt{n}} S_{\beta,n}(\beta_0, m_0) + o_p(1).$$

Hence,

$$\sqrt{n}(\hat{\beta} - \beta_0) \xrightarrow{d} N\{0, I(\beta_0)^{-1} \Sigma(\beta_0) I(\beta_0)^{-1}\}.$$

Under the information identity,

$$\Sigma(\beta_0) = I(\beta_0),$$

and therefore

$$\sqrt{n}(\hat{\beta} - \beta_0) \xrightarrow{d} N\{0, I(\beta_0)^{-1}\}.$$

This completes the proof. □

A.7 Proof of Theorem 3.7

Proof. The result follows by examining the three components that determine the stability and consistency of the sieve estimator: the approximation error, the stochastic estimation error, and the truncation-adjusted likelihood contribution. Recall from Theorem 3.6 that the estimation error of the nonlinear component admits the decomposition

$$\|\hat{m} - m_0\| = O_p \left(K_n^{-r} + \sqrt{\frac{K_n}{n}} + \mathcal{R}_{n,\lambda} \right),$$

where K_n^{-r} is the spline approximation error, $\sqrt{K_n/n}$ is the stochastic estimation error associated with the increasing sieve dimension, and $\mathcal{R}_{n,\lambda}$ denotes the additional contribution induced by penalization and the truncation-adjusted likelihood. We show that each of the conditions stated in the theorem is necessary for these terms to remain asymptotically negligible.

Case 1: $K_n/n \not\rightarrow 0$.

Suppose first that

$$\frac{K_n}{n} \not\rightarrow 0.$$

Then there exists a subsequence, denoted again by n , and a constant $c > 0$ such that

$$\frac{K_n}{n} \geq c$$

along that subsequence. Consequently,

$$\sqrt{\frac{K_n}{n}} \geq \sqrt{c} > 0.$$

Hence the stochastic component of the sieve estimation error does not converge to zero. More precisely, the empirical-process contribution to the criterion satisfies

$$\sup_{\theta \in \Theta_{K_n}} |(\mathbb{P}_n - P) \{\ell_\theta - \ell_{\theta_0}\}| = O_p \left(\sqrt{\frac{K_n}{n}} \right),$$

where $\mathbb{P}_n$ denotes the empirical measure and P the corresponding population expectation. Since the right-hand side is bounded away from zero, the stochastic perturbation of the population criterion does not disappear.

On the other hand, the spline approximation component satisfies

$$\inf_{m_{K_n} \in \mathcal{M}_{K_n}} \|m_{K_n} - m_0\| = O(K_n^{-r}),$$

which tends to zero whenever $K_n \rightarrow \infty$. Thus increasing the sieve dimension can reduce the approximation bias, but if K_n grows at a rate comparable to or faster than n , the reduction in approximation error is offset by the increasing stochastic variability.

Indeed, from the basic error decomposition,

$$\|\hat{m} - m_0\| \geq \text{stochastic contribution} - \text{remaining terms},$$

and the stochastic contribution cannot vanish when $K_n/n \not\rightarrow 0$. Therefore,

$$\|\hat{m} - m_0\| \not\rightarrow_p 0.$$

Thus the condition

$$\frac{K_n}{n} \rightarrow 0$$

is required to ensure that the effective dimension of the sieve does not increase too rapidly relative to the available sample size.

Case 2: $n\lambda_n \not\rightarrow \infty$.

Consider next the penalty parameter λ_n . The penalized objective function can be written schematically as

$$Q_n(\theta) = -\ell_n(\theta) + \lambda_n \theta^\top \Omega_n \theta,$$

where Ω_n is the spline penalty matrix. Its Hessian therefore has the form

$$\nabla^2 Q_n(\theta) = -\nabla^2 \ell_n(\theta) + 2\lambda_n \Omega_n.$$

The first component represents the information supplied by the observed data, whereas the second component provides regularization of the spline coefficients.

Suppose that

$$n\lambda_n \not\rightarrow \infty.$$

Then there exists a subsequence for which $n\lambda_n$ is bounded. Hence the regularization contribution

$$2\lambda_n \Omega_n$$

does not increase sufficiently fast relative to the stochastic variation associated with the increasing-dimensional parameter vector.

Let θ_{K_n} denote the best spline approximation to the true parameter. A local expansion of the penalized criterion around θ_{K_n} gives

$$0 = \nabla Q_n(\hat{\theta}) = \nabla Q_n(\theta_{K_n}) + \nabla^2 Q_n(\tilde{\theta})(\hat{\theta} - \theta_{K_n}),$$

for some intermediate point $\tilde{\theta}$. Therefore,

$$\hat{\theta} - \theta_{K_n} = -\left\{\nabla^2 Q_n(\tilde{\theta})\right\}^{-1} \nabla Q_n(\theta_{K_n}).$$

The stochastic score term is of order

$$O_p\left(\sqrt{\frac{K_n}{n}}\right),$$

while the penalty contributes the deterministic term

$$2\lambda_n \Omega_n \theta_{K_n}.$$

If $n\lambda_n$ does not diverge, the penalty is insufficient to provide the uniform curvature required to control the increasingly large spline parameter vector. Consequently, the minimum eigenvalue of the penalized Hessian cannot be guaranteed to remain bounded away from zero:

$$\inf_{\theta \in \Theta_{K_n}} \lambda_{\min} \{ \nabla^2 Q_n(\theta) \} \not\geq c_0 > 0.$$

The inverse Hessian can therefore become arbitrarily large, and the stochastic error

$$\|\hat{\theta} - \theta_{K_n}\| \leq \left\| \nabla^2 Q_n(\tilde{\theta})^{-1} \right\| \|\nabla Q_n(\theta_{K_n})\|$$

cannot be uniformly controlled.

As a consequence, the spline coefficient estimator may exhibit increasing variance and instability. Hence the estimation error of the nonlinear component does not necessarily converge to zero:

$$\|\hat{m} - m_0\| \not\rightarrow_p 0.$$

Therefore, the condition

$$n\lambda_n \rightarrow \infty$$

is required to maintain sufficient regularization and uniform control of the penalized criterion.

Case 3: $\inf_{x,z} C(x, z; \beta, m) = 0$.

Finally, consider the truncation-normalization term. The truncated conditional density is

$$f^{(T)}(y \mid x, z; \beta, m) = \frac{f_\varepsilon\{y - m(z) - g(x; \beta)\}}{C(x, z; \beta, m)},$$

where

$$C(x, z; \beta, m) = P(a_Y \leq Y \leq b_Y \mid X = x, Z = z).$$

Consequently, the log-likelihood contains the additional term

$$-\log C(x, z; \beta, m).$$

The corresponding parametric score therefore contains

$$-\frac{\partial}{\partial \beta} \log C(x, z; \beta, m) = -\frac{\partial C(x, z; \beta, m) / \partial \beta}{C(x, z; \beta, m)}.$$

Similarly, the derivative with respect to the nonlinear component contains terms of the form

$$-\frac{D_m C(x, z; \beta, m)[h]}{C(x, z; \beta, m)},$$

for perturbations h of m . Thus, unlike the ordinary untruncated likelihood, the score contains inverse powers of the normalization constant C .

Suppose now that

$$\inf_{x,z} C(x, z; \beta, m) = 0.$$

Then, for every $M > 0$, there exist values (x, z) such that

$$\frac{1}{C(x, z; \beta, m)} > M.$$

Therefore, the truncation-adjusted score contributions

$$\frac{\partial}{\partial \beta} \log C(X_i, Z_i; \beta, m)$$

can become arbitrarily large. The same phenomenon occurs for the functional derivative involving $D_m C$.

In particular, if the derivatives of C remain nonzero in regions where C approaches zero, then

$$\left| \frac{D_\beta C}{C} \right| \rightarrow \infty, \quad \left| \frac{D_m C[h]}{C} \right| \rightarrow \infty.$$

Consequently, the envelope required for uniform control of the empirical process,

$$\sup_{\theta \in \Theta_{K_n}} |(\mathbb{P}_n - P)\psi_\theta|,$$

where ψ_θ denotes the truncation-adjusted score contribution, need not be finite.

The same problem appears in the second derivative. Differentiating $\log C$ twice gives terms of the form

$$\frac{D^2 C}{C} - \frac{(DC)(DC)^\top}{C^2}.$$

Hence, as $C \rightarrow 0$, the Hessian contributions may diverge at rates of order C^{-1} or C^{-2} . The information matrix can consequently fail to be uniformly bounded and nonsingular. In particular, the regularity conditions required for the uniform law of large numbers and for the central limit theorem for the score are no longer guaranteed.

This demonstrates why the lower-bound condition

$$\inf_{x, z} C(x, z; \beta, m) \geq c_C > 0$$

is essential. It provides the uniform bound

$$\frac{1}{C(x, z; \beta, m)} \leq \frac{1}{c_C},$$

which in turn controls the truncation-adjusted score and its derivatives. It also prevents the likelihood normalization from amplifying stochastic fluctuations in regions where the probability of being observed is very small.

Conclusion: The three conditions control distinct components of the estimator. The condition $K_n/n \rightarrow 0$ controls the stochastic complexity of the growing sieve, the condition $n\lambda_n \rightarrow \infty$ guarantees sufficient regularization and curvature of the penalized criterion, and the condition

$$\inf_{x, z} C(x, z; \beta, m) > 0$$

controls the additional stochastic terms generated by the truncation-dependent normalization.

When these conditions hold, the approximation error K_n^{-r} , the stochastic error $\sqrt{K_n/n}$, and the penalization/truncation contributions can be controlled uniformly, which allows the consistency result established in Theorem 3.6 to hold. If any one of the conditions fails, at least one of the required components is no longer asymptotically negligible or uniformly controllable. Consequently, the convergence

$$\|\hat{m} - m_0\| \rightarrow_p 0$$

cannot, in general, be guaranteed.

□